CS 451/551 Midterm 2

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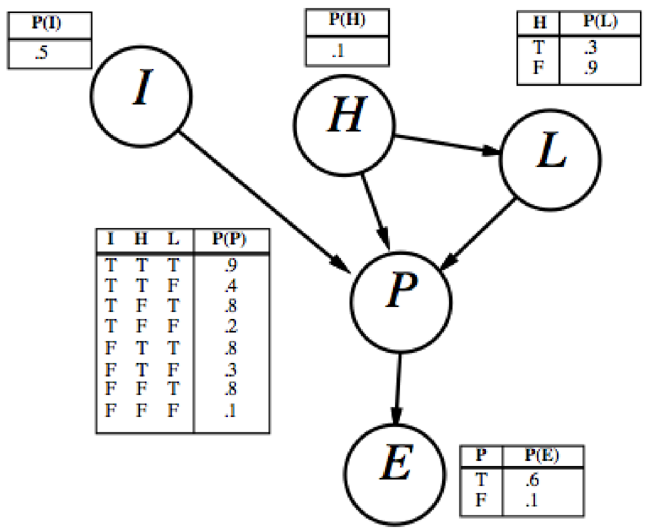
I. Question

Consider the simple Bayes net with boolean variables I = Intelligent, H = honest, P = popular, L = LotsOfCampaignFunds, E = Elected.



a. Is following are asserted by the network structure? Why? $P(I, L) = P(I)P(L)$
b. Is following are asserted by the network structure? Why? $P(E|P, L) = P(E|P, L, H)$
c. Is following are asserted by the network structure? Why? $P(P|I, H) \neq P(P|I, H, L)$
d. Calculate the value of P(i, h, ¬ l, p, ¬e).
e. Calculate the probability that someone is elected given that they are honest, have few campaign funds, and is intelligent.

II. Question

Given the Bayes Net in Question 1, sample 10 points from the network with

a. Rejection sampling
b. Likelihood weighting

to calculate the probability that someone is elected given that they are honest, have few campaign funds, and is intelligent.

III. Question

a. Consider a single Boolean random variable Y (The 'classification'). Let the prior probability P(Y = true) be π. Let's try to find π given a training set $D = (y_1, y_2, \ldots, y_N)$ with N independent samples of Y. Furthermore, suppose p of the N are positive and n of the N are negative. Write down an expression for the likelihood of D (the probability of seeing this particular sequence of examples, given a fixed value of π) in terms of π, p, and n.
b. By differentiating the loglikelihood L, find the value of π that maximizes the likelihood.
c. Now suppose we add in k Boolean random variables $X_1, X_2, \ldots, X_k$ (the 'attributes') that describes each sample, and suppose we assume that the attributes are conditionally independent of each other given the goal Y. Draw the Bayes net corresponding to this assumption.
d. Write down the likelihood for the data including the attributes, using the following additional assumption:

- α_i is $P(X_i = true|Y = true)$
- β_i is $P(X_i = true|Y = false)$
- p_i^+ is the count of samples for which X_i = true, Y = true.
- n_i^+ is the count of samples for which X_i = false, Y = true.
- p_i^- is the count of samples for which X_i = true, Y = false.
- n_i^- is the count of samples for which X_i = false, Y = false.

[Hint: consider first the probability of seeing a single example with specified values for $X_1, X_2, \ldots, X_k, Y$]

e. By differentiating the loglikelihood L, find the values α_i and β_i (in terms of various counts) that maximize the likelihood and say in words what these values represent.
f. Let k = 2, and consider a dataset with four examples as follows:

X_1	X_2	Y
0	0	0
0	1	1
1	0	1
1	1	0

Compute the maximum likelihood estimates of $\pi, \alpha_1, \alpha_2, \beta_1, \beta_2$.

g. Given these estimates of $\pi, \alpha_1, \alpha_2, \beta_1, \beta_2$, what are the posterior probabilities $P(Y = true|X_1, X_2)$ for each example.

IV. Question

Consider learning a decision tree that you could use to judge whether or not you will like a given restaurant. Assume you have chosen to use the following three features to describe restaurants, with the possible values shown.

Price $\in$ {Low, Med, High} Type $\in$ {Hamburgers, Pizza, Fish, Vegetarian}

Assume Quinlan's ID3 algorithm is given the following set of classified training examples. Calculate the decision tree that ID3 would produce. Show all your work. (You may use the abbreviations that are used to describe the examples.)

P	T	Like
L	H	+
L	V	+
M	F	-
M	V	+
H	P	-

$\log 2 = 1, \log 3 = 1.58, \log 4 = 2, \log 5 = 2.32.$

V. Question

a. Draw a suitable network topology for the following set of two-valued nodes **FrozenBattery**, **IcyWeather**, **CarWontStart**, **NoGas**.
b. Give reasonable conditional probability tables associated with the **FrozenBattery** and **CarWontStart** nodes in your network.
c. How many independent values are contained in the joint probability distribution for four two-valued nodes assuming no conditional independence relations are known to hold among them.
d. How many independent probability values do your network tables contain include priors as appropriate?
e. Suppose we add another node for a broken starter motor called **BrokenSM**. Where would it go? Say briefly how your existing table for **CarWontStart** should be changed?